P-256 Solinas z-score recalibration

P-256 Solinas bit-pair correlations: z-score recalibration (2026-10-22)

Defect

`flag_outliers` and the test `solinas_correlation_z_distribution` in `src/cryptanalysis/solinas_correlations.rs` computed `z = (p_joint - E) / sqrt(E(1-E)/n)` with `E = p_r * p_c`. Under independence with estimated marginals the null standard error of `p_joint - p_r p_c` is `sqrt(p_r(1-p_r) p_c(1-p_c) / n)`. At `p_r = p_c = 1/2` the old scale is `sqrt(0.1875/n)` against the correct `sqrt(0.0625/n)`, so every z was too small by `sqrt(3)`. The fix is `CorrelationTable::z_score`, used by both callers.

Calibration test

`z_score_is_calibrated_on_independent_bits` (seed 20261022): 64 x 128 = 8,192 pairs of independent uniform bits, n = 4,000. Result (`calibration_test_output.txt`): fraction |z| > 3 is 0.00183 (nominal 0.0027), max |z| 3.45 (sqrt(2 ln m) = 4.25), E[z^2] within 0.9 to 1.1 (asserted). The test asserts fraction in [0.0008, 0.0060], max within 0.8 of sqrt(2 ln m), and E[z^2] in [0.9, 1.1]. The old scale gives fraction about 0 and max near 2.0 on this population. The pairs share marginals, so they are not independent and the max sits below the independent-pairs value; the test bounds are set for that.

Re-run

```
SOLINAS_CORR_N=1000000 cargo test --release --lib \

cryptanalysis::solinas_correlations::tests::solinas_micro_bit_correlat:
\
  -- --nocapture --exact
```

Seed 42 (fixed in the test), N = 10^6, release build, shared target dir. Raw output: `raw_output_n1e6.txt`. The test's own runtime is 75.2 s (the `real` time of 27 min 58 s in the file includes waiting for the shared cargo lock and the build).

	old scale (recorded)	corrected
max |z|	2.46	4.26
top 20 range	2.17 to 2.46	3.76 to 4.26
Bonferroni threshold (alpha 0.001/131,072, as the test prints)	6.11	6.11

Ratio 4.26 / 2.46 = 1.73, the predicted sqrt(3). Top outliers (|z|): (149, 54) 4.26, (197, 164) 4.12, (117, 119) 4.02, (103, 101) 4.01, (122, 185) 3.97. Pairs are (output bit i, input bit j).

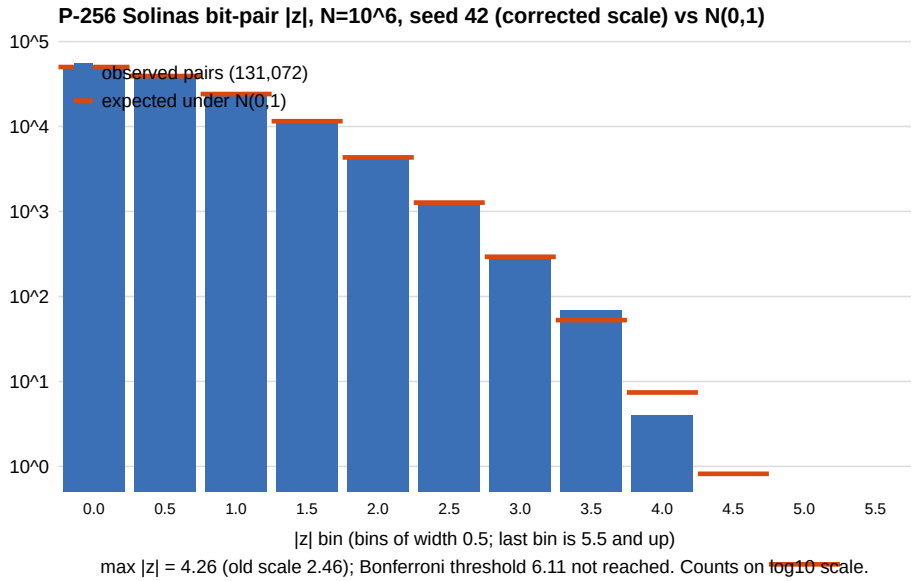
The histogram of all 131,072 |z| (`z_distribution.svg`, counts in `expected.tsv`) matches the standard normal: bins of width 0.5 observed versus expected 49,972/50,191; 39,615/39,291; 23,998/24,077; 11,484/11,549; 4,392/4,336; 1,263/1,274; 276/293; 68/53; 4/7.4; 0/0.8.

Verdict

The null verdict survives. The corrected max |z| = 4.26 is the size a calibrated null gives for 131,072 pairs (sqrt(2 ln m) = 4.84 for m = 131,072), and it is far below the 6.11 threshold. The old recorded value 2.46 was an artefact of the deflated scale, not evidence for an unusually clean null. This is a single-seed run; it is not an independent replication.

Files

- `raw_output_n1e6.txt`, `calibration_test_output.txt`: raw command output.
- `z_distribution.svg`: |z| histogram versus N(0,1); `expected.tsv`: bin counts.
- `report.pdf`: this README rendered (pandoc, then headless Chromium).



|z| distribution versus N(0,1)